1 Solution — GIAM 8.3 Problem 3

1.1 Problem

In the proof of Cantor's theorem we construct a set S that cannot be in the image of a presumed bijection from A to $\mathcal{P}(A)$. Suppose $A = \{1, 2, 3\}$ and f determines the following correspondences:

$$1 \longleftrightarrow \emptyset, \qquad 2 \longleftrightarrow \{1, 3\}, \qquad 3 \longleftrightarrow \{1, 2, 3\}.$$

What is S?

1.2 Cantor's Recipe for S

In the proof of Cantor's theorem (Theorem 8.3.1 in the textbook), given any function $f : A \to \mathcal{P}(A)$, define

$$S = \{ x \in A : x \notin f(x) \}.$$

The argument: $S \subseteq A$, so $S \in \mathcal{P}(A)$. If f were surjective, some $y \in A$ would have $f(y) = S$, but then "$y \in S$?" leads to a contradiction either way (see the §8.3 main proof). So S is necessarily *not* in the image of f — establishing that no $f : A \to \mathcal{P}(A)$ can be surjective, i.e., $|A| < |\mathcal{P}(A)|$.

For the given f, computing S amounts to checking, for each $x \in A$, the single yes/no question *"is $x \in f(x)$?"* and collecting the x's for which the answer is **no**.

1.3 Computing S Element-by-Element

- $x = 1$: $f(1) = \emptyset$. Is $1 \in \emptyset$? **No**. So $1 \in S$.
- $x = 2$: $f(2) = \{1, 3\}$. Is $2 \in \{1, 3\}$? **No**. So $2 \in S$.
- $x = 3$: $f(3) = \{1, 2, 3\}$. Is $3 \in \{1, 2, 3\}$? **Yes**. So $3 \notin S$.

Therefore

$$\boxed{S \;=\; \{1,\,2\}}.$$

1.4 The Diagonal Membership Table

The same computation, laid out as a membership table — exactly the "table whose diagonal we flip" view of Cantor's construction:

Cantor Diagonal Membership Table

For each $x \in A$: is $x \in f(x)$? Flip the diagonal answers to build S.

f(x)	1	2	3	x ∈ f(x)?	x ∈ S?
f(1) = ∅	✗	✗	✗	1 ∈ ∅? **No**	**Yes → 1 ∈ S**
f(2) = {1, 3}	✓	✗	✓	2 ∈ {1,3}? **No**	**Yes → 2 ∈ S**
f(3) = {1, 2, 3}	✓	✓	✓	3 ∈ {1,2,3}? **Yes**	No (3 ∉ S)

S = {x ∈ A : x ∉ f(x)} = {1, 2}

Figure 1.1: Cantor diagonal membership table. The diagonal cells (yellow) answer 'is x in f(x)?'; flipping each answer gives 'is x in S?'. Here the diagonal answers are No, No, Yes — so S contains 1 and 2 but not 3.

The yellow cells form the diagonal. *Flipping* the diagonal — replacing each "yes" with a "no" and vice versa — gives the membership table of S. This is exactly the diagonal-flip trick used in the §8.3 Problem 1 / 2 arguments for $(0,1)$, transferred from "decimal digits" to "yes/no membership bits."

1.5 Verifying S Is Missed by f

The image of f is

$$\operatorname{image}(f) \;=\; \{\, f(1),\ f(2),\ f(3)\,\} \;=\; \{\,\emptyset,\ \{1,3\},\ \{1,2,3\}\,\}.$$

Is $S = \{1,2\}$ in this image? Comparing:

- $\{1, 2\} \neq \emptyset$ ✓
- $\{1, 2\} \neq \{1, 3\}$ (differs at 2 and 3) ✓
- $\{1, 2\} \neq \{1, 2, 3\}$ (differs at 3) ✓

So $S \notin \text{image}(f)$. Cantor's diagonal construction has done its job: it has *exhibited* a subset of A that f misses.

1.6 Powerset View

There are $|\mathcal{P}(A)| = 2^3 = 8$ subsets of A in total. The image of f covers 3 of them; 5 are missed. $S = \{1, 2\}$ is one of those 5:

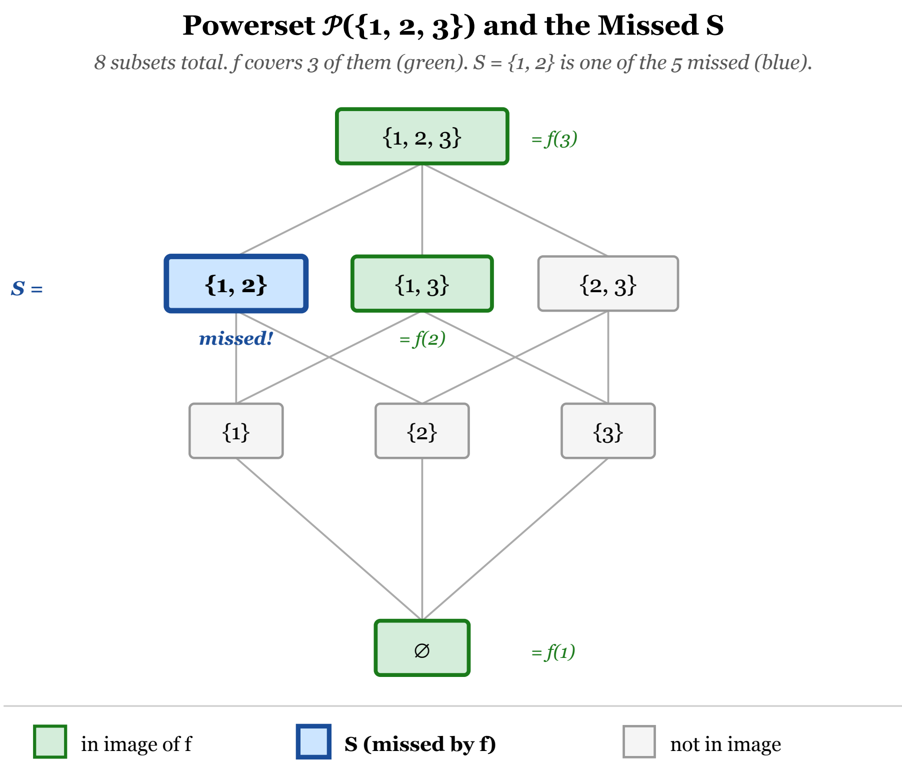

Figure 1.2: All 8 subsets of A = {1, 2, 3}. The image of f covers ∅, {1,3}, and {1,2,3} (green). The diagonal set S = {1, 2} (blue) sits among the four subsets of A that f misses.

1.7 How the Diagonal Argument Works in General

The set $S = \{x \in A : x \notin f(x)\}$ is designed to be different from *every* $f(y)$. For each $y \in A$, S disagrees with $f(y)$ at the element y itself:

- If $y \in f(y)$, then $y \notin S$ (so S excludes y while $f(y)$ includes it).
- If $y \notin f(y)$, then $y \in S$ (so S includes y while $f(y)$ excludes it).

In both cases, S differs from $f(y)$ at element y, so $S \neq f(y)$. Since this holds for every $y \in A$, $S \notin \textbf{image}(f)$.

This is the *exact* structural analog of the decimal/ternary diagonalization arguments (Problems 1 and 2 of this section): there, we built a real r differing from each r_n at the n-th *digit*; here, we build a set S differing from each $f(y)$ at the element y. The membership-bit version (here) is in fact the cleanest and most general — it works for any set A, finite or infinite, and doesn't have to worry about dual-representation issues.

1.8 Remark — f Wasn't a Bijection Anyway

The problem says "presumed bijection," but the given f is *manifestly not* a bijection: it maps $|A| = 3$ elements to only 3 of the $2^3 = 8$ subsets, so it isn't surjective. Cantor's theorem reassures us that this *had* to happen — there is no surjection $A \to \mathcal{P}(A)$ for any finite set A (since $2^n > n$ for all $n \geq 0$), let alone for the deeper case of infinite A.

So the diagonal construction in this finite example is doing two things at once:

1. *Reminding us* of the gap between $|A| = 3$ and $|\mathcal{P}(A)| = 8$ — there are 5 "spare" subsets that no map from a 3-element set could possibly cover.
2. *Picking out one specific* missed subset, $S = \{1, 2\}$, generated by the diagonal flip.

For an infinite A, claim (1) is non-trivial — finite cardinality counting no longer applies — and Cantor's diagonal argument is the only known way to establish it. For finite A, the argument is just a clever demonstration of the inevitable.

1.9 Remark — Why S Depends on the Choice of f

Different functions f produce different sets S. For instance:

- The given f (with $f(1) = \emptyset$, $f(2) = \{1, 3\}$, $f(3) = \{1, 2, 3\}$) yields $S = \{1, 2\}$.

- The "identity-ish" map $f(x) = \{x\}$ (i.e., $1 \mapsto \{1\}$, $2 \mapsto \{2\}$, $3 \mapsto \{3\}$) has $x \in f(x)$ for every x, so $S = \emptyset$. (S is then $f(\text{nobody})$, but $S = \emptyset$ is *also* not in the image of this f, since $\emptyset \neq \{x\}$ for any x.)
- The map $f(x) = A \setminus \{x\}$ (i.e., $1 \mapsto \{2,3\}$, $2 \mapsto \{1,3\}$, $3 \mapsto \{1,2\}$) has $x \notin f(x)$ for every x, so $S = \{1,2,3\} = A$.

In every case, the constructed S is missed by the corresponding f, but *which* set S depends on which f we started from. The diagonal recipe is a uniform procedure — apply it to any f and you get a guaranteed-missed subset.

1.10 Remark — Could We "Patch" f to Include S?

Suppose we tried to repair f by reassigning some element of A to map to $S = \{1,2\}$. E.g., redefine $f'(2) = \{1,2\}$. Now f''s image includes $S = \{1,2\}$ — but the *new* function f' has a *new* diagonal set S', which we can recompute:

- $1 \in f'(1) = \emptyset$? No. So $1 \in S'$.
- $2 \in f'(2) = \{1,2\}$? Yes. So $2 \notin S'$.
- $3 \in f'(3) = \{1,2,3\}$? Yes. So $3 \notin S'$.

So $S' = \{1\}$, and this *new* S' is now missed by f' (check: $\{1\} \notin \{\emptyset, \{1,2\}, \{1,2,3\}\}$).

The patching shifts the missed subset around but never eliminates it — every function $A \to \mathcal{P}(A)$ has *some* diagonal subset S outside its image. This is the structural inevitability that Cantor's theorem captures: $\mathcal{P}(A)$ is genuinely too big for A to enumerate.